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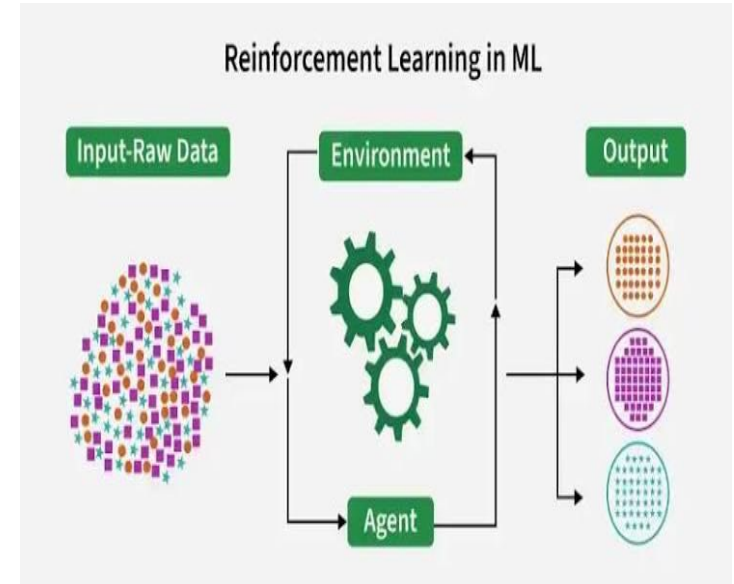
Dynamic Programming for Sequential Decision- Making in Reinforcement Learning

Guided BY
Dr.A.SENTHIL

PRESENTED BY
K.Neeha (192425245)

INTRODUCTION

- What is Dynamic Programming?
- Breaks complex problems into smaller subproblems
- Used in Reinforcement Learning
- Finds optimal decisions



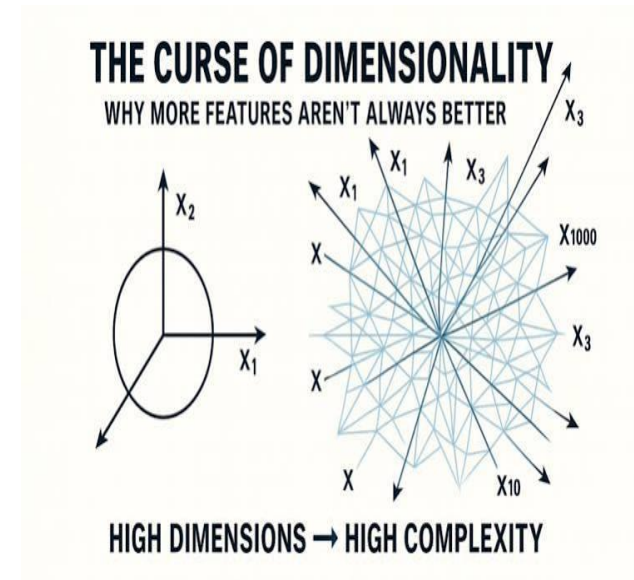
Advantages

- Strong mathematical foundation
- Reduces computational complexity through subproblem reuse
- Ensures convergence to the optimal policy under suitable conditions
- Widely applicable in planning and control problems



Limitations

- Requires a complete environment model
- Suffers from the curse of dimensionality
- High computational cost for large state spaces
- Requires significant memory storage
- Not suitable for very complex real-world environments without approximations
- Can be slow when the number of states and actions is very large



Conclusion

- Bellman Principle, Policy Evaluation, Policy Improvement,
- Value Iteration and Policy Iteration are key RL methods.
- Dynamic Programming is a fundamental technique in Reinforcement Learning for solving sequential decision-making problems.
- The Bellman Principle of Optimality provides the foundation for finding optimal policies.

THANK YOU